Kavya Raj

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PROFESSIONAL PROFILE

Android Software Engineer with **5+ years of experience** building and maintaining enterprise Android applications using **Kotlin, Jetpack Compose, MVVM, Clean Architecture, and REST APIs**. Skilled in developing scalable mobile solutions, integrating **third-party SDKs**, implementing **CI/CD practices**, and delivering production-ready applications. Experienced across the full software development lifecycle, from architecture and development to testing, deployment, and support. **MSc Artificial Intelligence and Robotics** graduate with additional expertise in **Flutter and AI-powered application** development.

TECHNICAL SKILLS

• **Languages & Core:** Kotlin (Primary), Java, Dart, Python, SQL, XML, JSON
• **Modern Android Development:** Jetpack Compose, Composables, State Management, Recomposition, Compose Navigation, Material 3, Adaptive Layouts, Compose Animations
• **Architecture & Design:** MVVM, MVI, Clean Architecture, Repository Pattern, Use Case Layer, Single Activity Architecture, Multi-Module Architecture, SOLID Principles, Reusable Software Components, Modular Architecture, Object-Oriented Programming, Design Patterns
• **Jetpack & AndroidX:** ViewModel, StateFlow, LiveData, Room Database, Paging 3, WorkManager, Navigation Component, CameraX, Data Binding
• **Dependency Injection:** Hilt, Dagger 2
• **Asynchronous & Reactive:** Kotlin Coroutines, Kotlin Flow, StateFlow, SharedFlow, RxJava (Legacy Maintenance)
• **Networking & APIs:** Retrofit, OkHttp, REST APIs, JSON Parsing, API Integration, Microservice API Integration, Client-Server Communication, OAuth 2.0, JWT, SSL Certificate Pinning
• **Local Data & Persistence:** Room Database, SQLite, SQL, DataStore, EncryptedSharedPreferences, Offline Data Management, SharedPreferences
• Firebase & Cloud: Firebase Authentication, Firestore, Firebase Cloud Messaging (FCM), Crashlytics, Google Cloud Platform (Basic Exposure), Azure, Azure DevOps
• **Testing & Quality:** JUnit, Espresso, Unit Testing, Integration Testing, Instrumented Testing, Test-Driven Development (TDD), Code Reviews, Debugging, Root Cause Analysis, Performance Optimisation, Application Monitoring
• **Security:** Android Security Best Practices, Android Keystore System, OAuth 2.0, JWT, SSL Certificate Pinning, Secure API Communication
• **Build, CI/CD & Release:** Gradle, Build Variants, Product Flavours, Git, GitHub, GitHub Actions, Jenkins, GitLab CI/CD, Azure DevOps, Continuous Integration, Continuous Deployment, Build Automation, Release Management, Google Play Console (Internal → Production), App Deployment
• **Device Integration & Connectivity:** Sunmi POS SDK, Epson Printer SDK, Bluetooth Printer SDKs, Thermal Printers, Barcode Scanners, Hardware SDK Integration, Bluetooth Connectivity
• **Cross-Platform Development:** Flutter, Dart, Responsive Mobile Development, Gemini AI Integration
• **AI & Machine Learning**: Machine Learning, Neural Networks, Anomaly Detection, Classification Models, Feature Engineering, Data Preprocessing, Model Evaluation, Privacy-Preserving Machine Learning, Federated Learning Concepts, Differential Privacy Concepts
• **Collaboration & Delivery:** Agile Development, Scrum, Sprint Planning, Sprint Delivery, Stakeholder Collaboration, Cross-Functional Team Working, Technical Documentation, Mentoring, Product Development Lifecycle, Software Development Lifecycle (SDLC)

EXPERIENCE

Android Developer

Synergia Software IT Infrastructure LLC **January 2022 - August 2024, Dubai, UAE**

- Developed and maintained enterprise Android ERP and POS applications using Kotlin, Jetpack Compose, MVVM, MVI, and Clean Architecture, supporting thousands of users across multiple business deployments.
- Designed scalable application architecture and reusable software components, improving code maintainability by 30% and accelerating feature development.
- Delivered 20+ production-ready features using modern Android development practices, aligning with complex business workflows and operational requirements.
- Built modern user interfaces using Jetpack Compose, Material Design 3, Compose Navigation, and adaptive layouts to enhance user experience and usability.
- Utilized Kotlin Coroutines, Flow, StateFlow, and Hilt to implement scalable, maintainable, and testable application solutions.
- Integrated Android applications with microservice-based REST APIs and backend ERP services, improving data synchronization reliability by 25%.
- Developed reusable modules and shared components, reducing development effort by 25% and improving consistency across projects.
- Integrated third-party SDKs including POS device SDKs, Bluetooth printer SDKs, thermal printer solutions, and barcode scanner integrations.
- Optimized API communication, UI responsiveness, and application performance, reducing response latency by 25%.
- Performed code reviews, unit testing, integration testing, debugging, and root cause analysis, reducing application crash rates by 35%.
- Contributed to CI/CD pipelines, automated build processes, Google Play release activities, and deployment workflows, reducing deployment turnaround time by 20%.

- Collaborated with product managers, UX designers, QA engineers, backend developers, and business stakeholders to deliver high-impact product improvements.
- Mentored junior developers and contributed to architecture planning, technical decision-making, and engineering best practices.

Android Developer

Microtech Software Solutions **June 2019 - December 2021, Kerala, India**

- Developed Android applications using Java, Kotlin, MVVM architecture, and REST API integrations to support enterprise business operations.
- Designed and delivered 15+ production-ready features for ERP, financial, and business management applications aligned with client requirements.
- Developed backend APIs using ASP.NET and integrated Android applications with enterprise systems, improving system integration efficiency by 20%.
- Implemented reusable software components and modular application structures to improve maintainability and development productivity.
- Integrated secure REST APIs, authentication mechanisms, and backend services, improving data exchange reliability and application stability.
- Built and maintained SQL-based database operations, improving data retrieval performance by 25%.
- Participated in code reviews, software testing, debugging, and production issue resolution to improve software quality and release reliability.
- Collaborated with cross-functional teams including product stakeholders, designers, and developers to deliver customer-focused software solutions.
- Contributed to Agile sprint planning, release management activities, and continuous improvement initiatives across multiple development cycles.
- Supported architecture discussions and engineering improvements that enhanced scalability, maintainability, and long-term product sustainability.

KEY PROJECT

AI Career Coach Mobile Application | Flutter, Dart, Gemini AI

github.com/kavyarajtntd-a11y/AI-Career-Coach-Flutter-Gemini-AI

- Developed a cross-platform mobile application using Flutter and Dart.
- Integrated Gemini AI to provide AI-powered career guidance and interview preparation support.
- Implemented responsive user interfaces and AI-driven recommendation workflows.
- Consumed REST APIs and AI services to generate personalized career insights.
- Managed source code using Git and GitHub.

Data Protection Challenges in Information Security and application of AI in mitigating the issue using machine learning

University of Hertfordshire

- Designed and developed machine learning workflows in Python for anomaly detection and intelligent threat analysis in cloud-oriented environments.
- Applied exploratory data analysis, preprocessing, feature engineering, classification modelling, and evaluation metrics including accuracy, precision, recall, and F1 score.
- Explored privacy-preserving AI techniques including federated learning and differential privacy for secure model deployment.
- Built experimentation workflows using Python, Jupyter Notebook, and Google Colab.
- Investigated intelligent detection of abnormal system behaviour using machine learning-based threat modelling.

EDUCATION

MSc Artificial Intelligence and Robotics

University of Hertfordshire • Hatfield, United Kingdom • 2026

Bachelor of Computer Application

Mahatma Gandhi University • Kottayam, Kerala • 2019

PROFESSIONAL DEVELOPMENT

AI-Driven Data Analytics Programme (In Progress)

NeST Digital Academy / Entri Elevate • 2026

- Comprehensive training in Data Analytics, Power BI, MySQL, Python, and AI-driven decision-making.
- Developing practical skills in data cleaning, transformation, exploratory data analysis, reporting, and business intelligence.
- Working on assignments, mini projects, and a capstone project focused on real-world analytics applications.
- Gaining hands-on experience with data visualization, statistical analysis, and data-driven problem solving.
- Participating in industry-led mentoring, soft skills training, and project-based learning.

CERTIFICATIONS

NVIDIA Fundamentals of Deep Learning

Google AI Essentials